



Picture-In-Picture Overlay Workspace Utility

Documentation Report

Submitted By

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Course: Frontend Development Case Study Project

Student Details

Field	Information
Name	Pallavi Sarovar
Roll Number	150096725219
Project Title	Picture-In-Picture Overlay Workspace Utility

Technology	ReactJS, Vite, Zustand, Tailwind CSS
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Deployment	Vercel
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Problem Statement

Modern productivity workflows require users to continuously switch between browser tabs and applications. Important tools such as notes, timers, media streams, and task trackers often become hidden behind other windows, reducing productivity and focus.

The objective of this project is to create a frontend-only web application that allows users to detach workspace widgets into always-on-top floating windows using the Document Picture-in-Picture API. This enables users to continue interacting with important workspace tools while working in other applications.

Tech Stack

Frontend Technologies

- ReactJS
- Vite
- JavaScript (ES6+)
- Tailwind CSS

State Management

- Zustand

Browser APIs

- Document Picture-in-Picture API
- Picture-in-Picture API
- MediaDevices API
- Screen Capture API

- LocalStorage API

Deployment

- Vercel
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Component Architecture

Main Components

1. Dashboard

- Displays available workspace modules.
- Allows users to launch widgets.

2. FloatingNotesWidget

- Markdown note editor.
- Checklist management.
- Real-time synchronization.

3. StopwatchWidget

- Focus timer.
- Session tracking.
- Picture-in-Picture support.

4. ScreenShareWidget

- Webcam streaming.
- Screen sharing.
- Video management.

5. TelemetryConsole

- Event monitoring.
- Activity logging.
- System diagnostics.

6. workspaceStore.js

- Global state management.
- Synchronization between windows.
- Persistence handling.

7. usePictureInPicture.js

- Creates PiP windows.
 - Clones styles.
 - Handles PiP lifecycle events.
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Features

Workspace Management

- Multiple productivity widgets.
- Easy workspace switching.

Floating Picture-in-Picture Windows

- Always-on-top widgets.
- Independent floating windows.

Notes Management

- Markdown support.
- Checklist support.
- Automatic synchronization.

Focus Stopwatch

- Start, pause, and reset timer.
- Session tracking.

Media Stream Viewer

- Webcam support.
- Screen sharing.
- Video stream management.

Telemetry Monitoring

- Event logging.
- Workspace activity tracking.
- PiP lifecycle monitoring.

Persistence

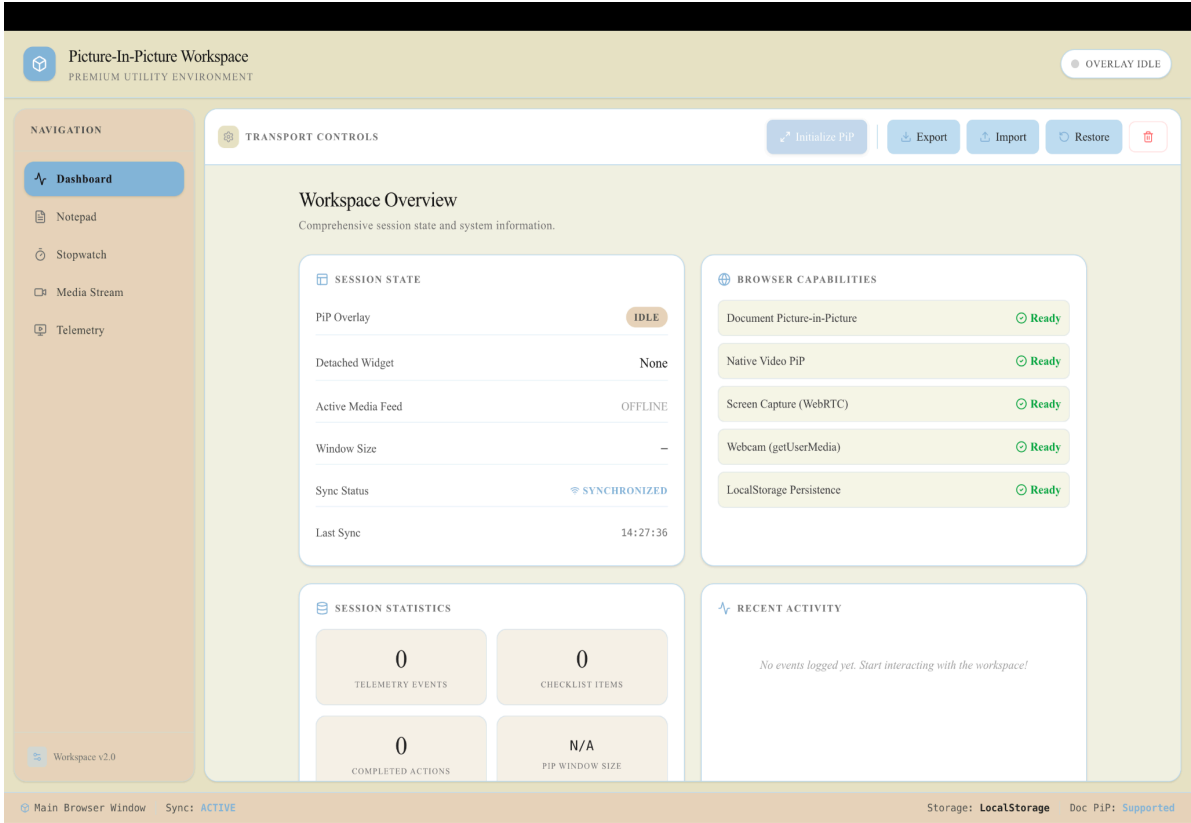
- LocalStorage-based data storage.
- Session recovery after refresh.

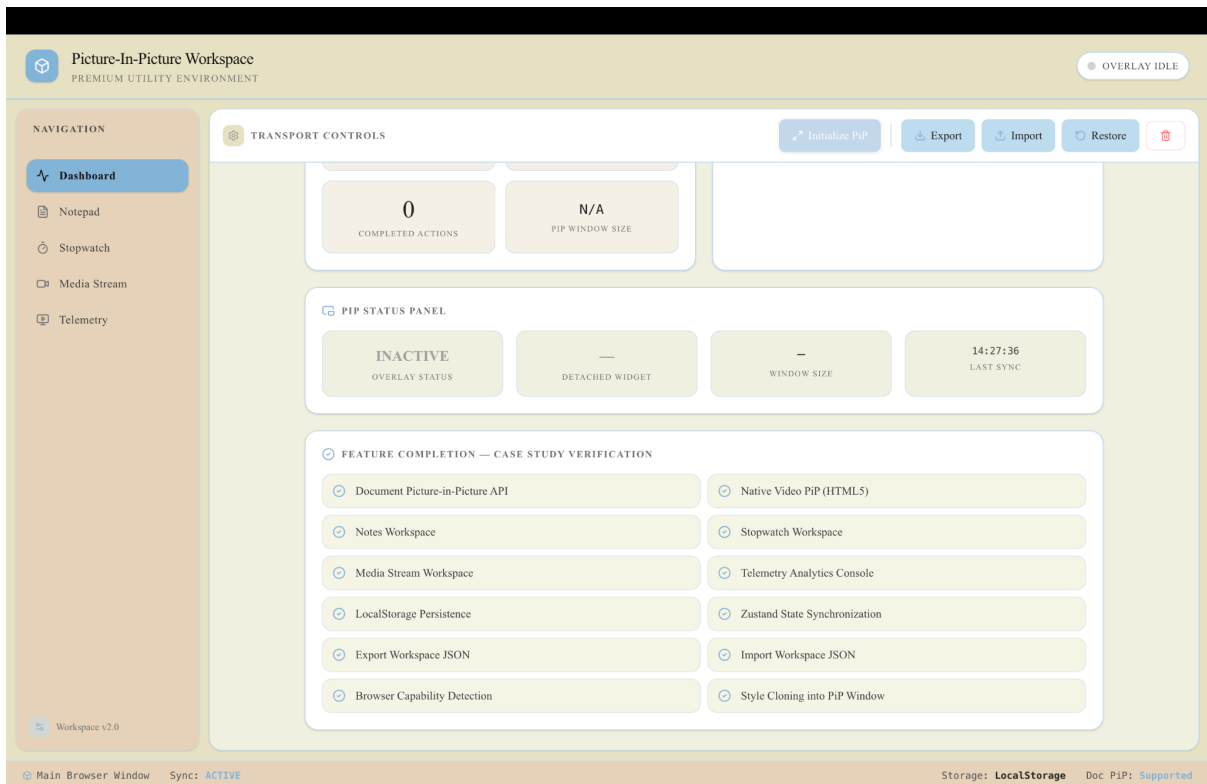
Import & Export

- Export workspace data as JSON.
- Restore previous configurations.

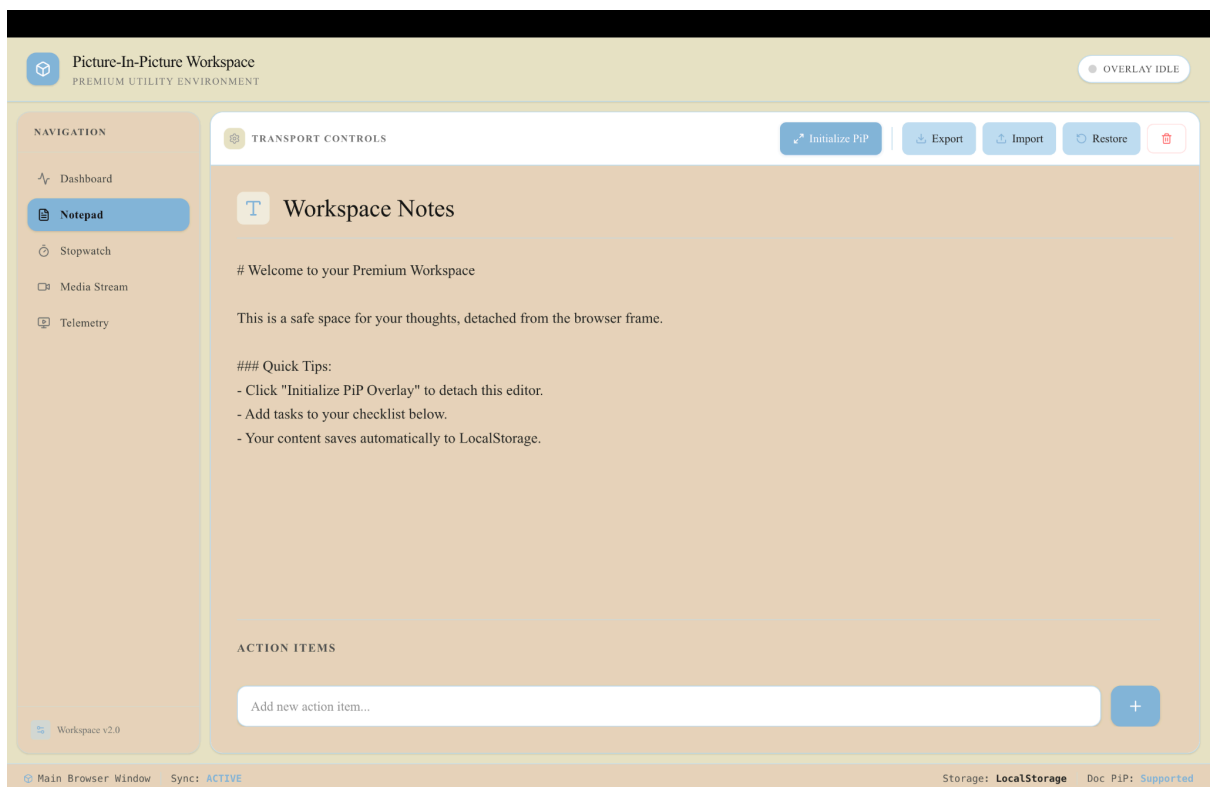
Screenshots

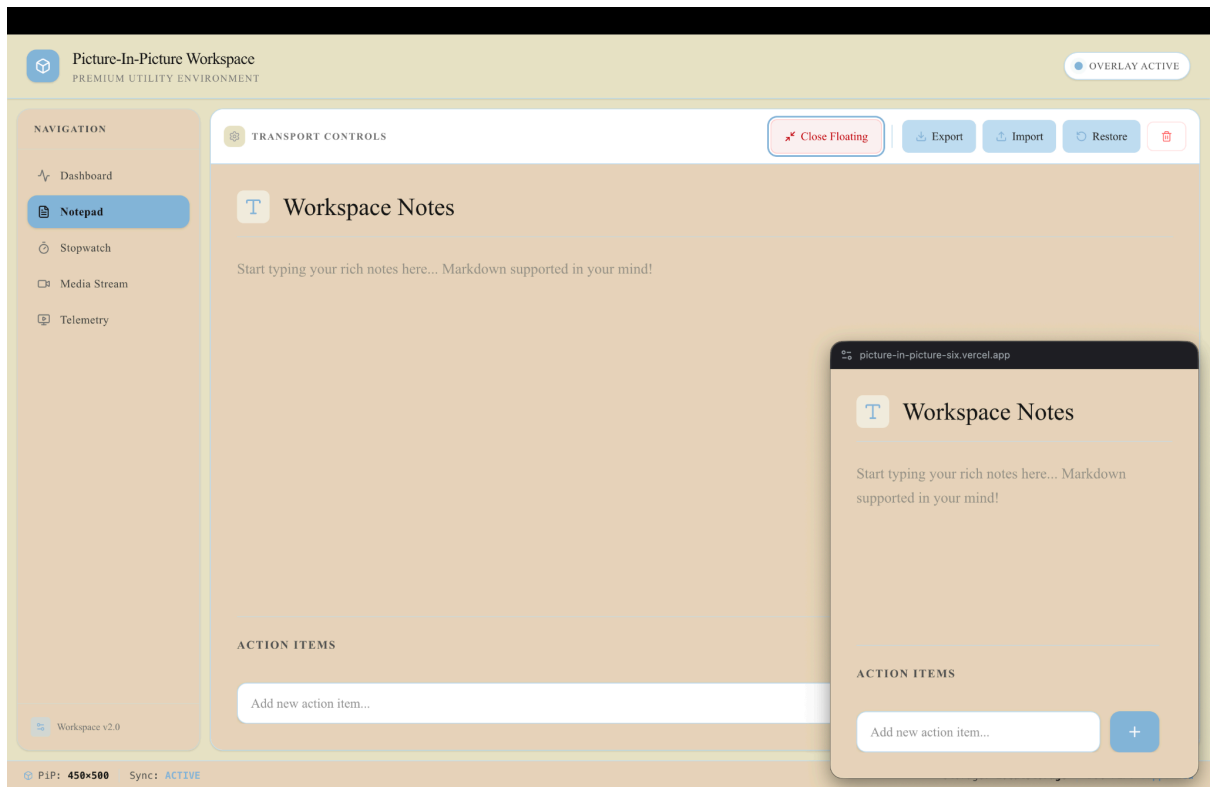
Screenshot 1 – Dashboard Interface



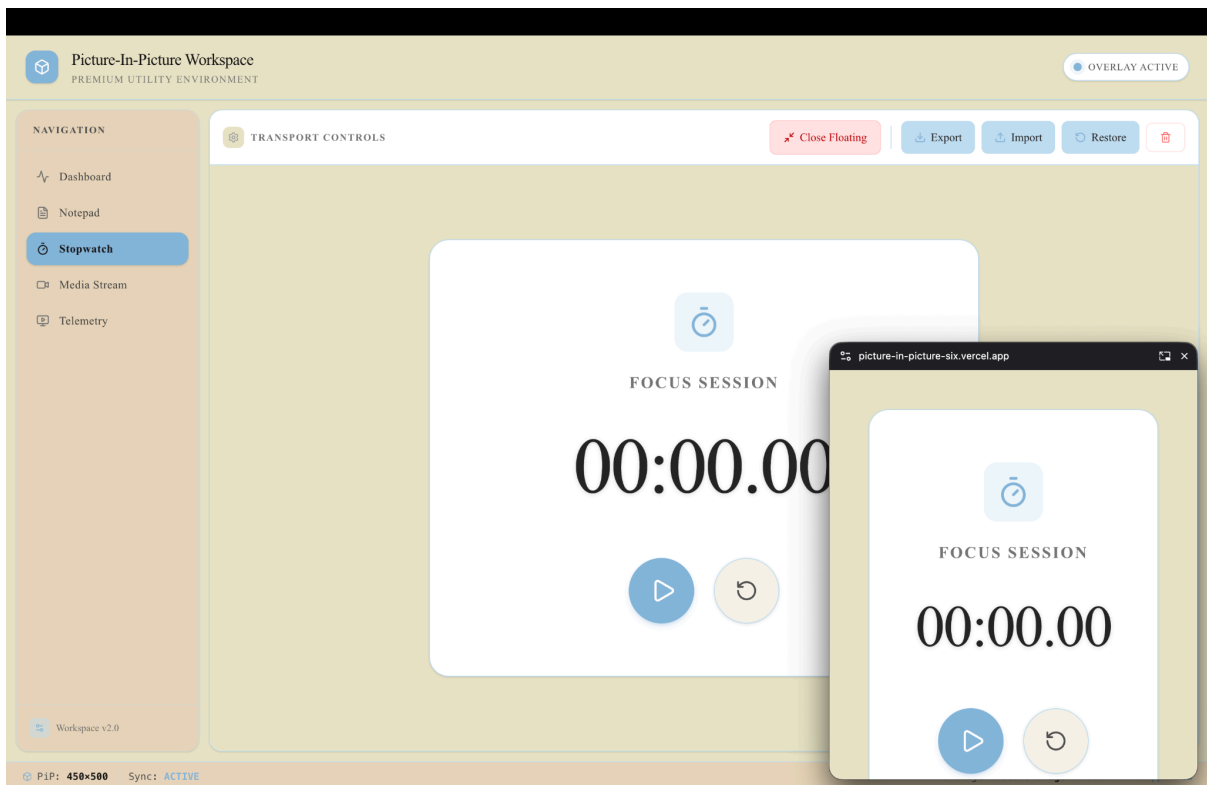


Screenshot 2 – Notes Workspace

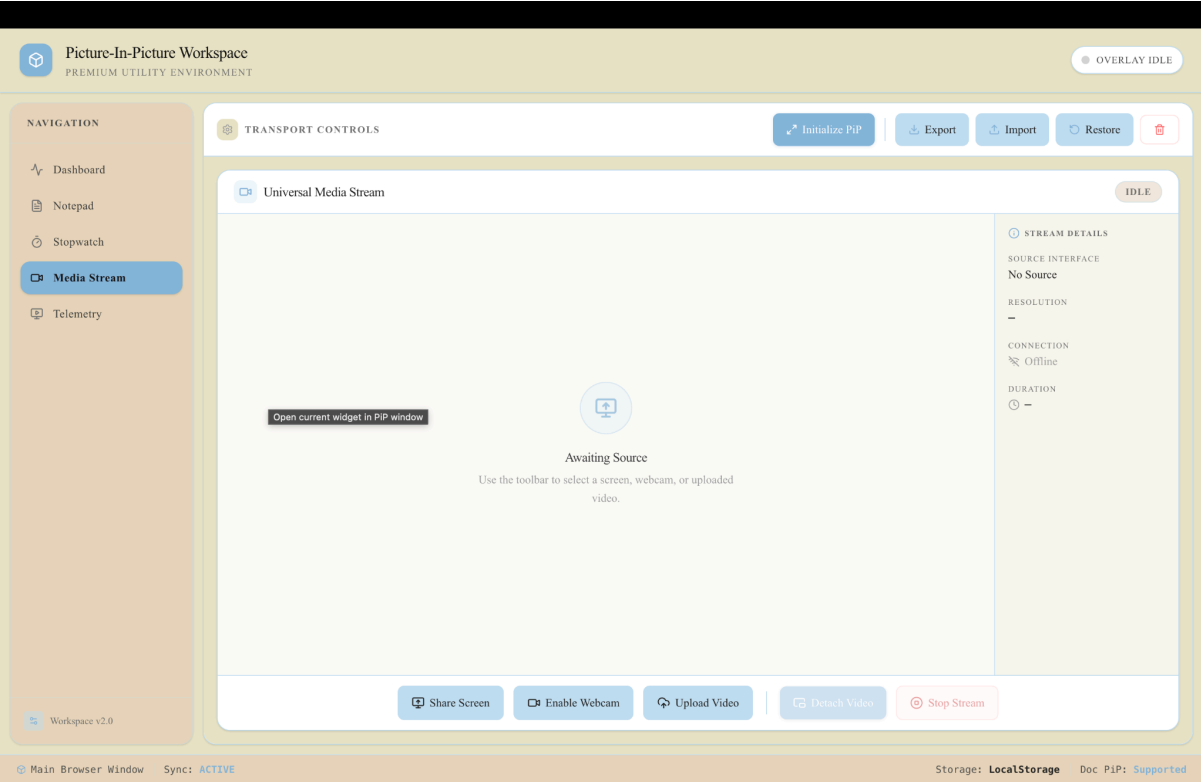




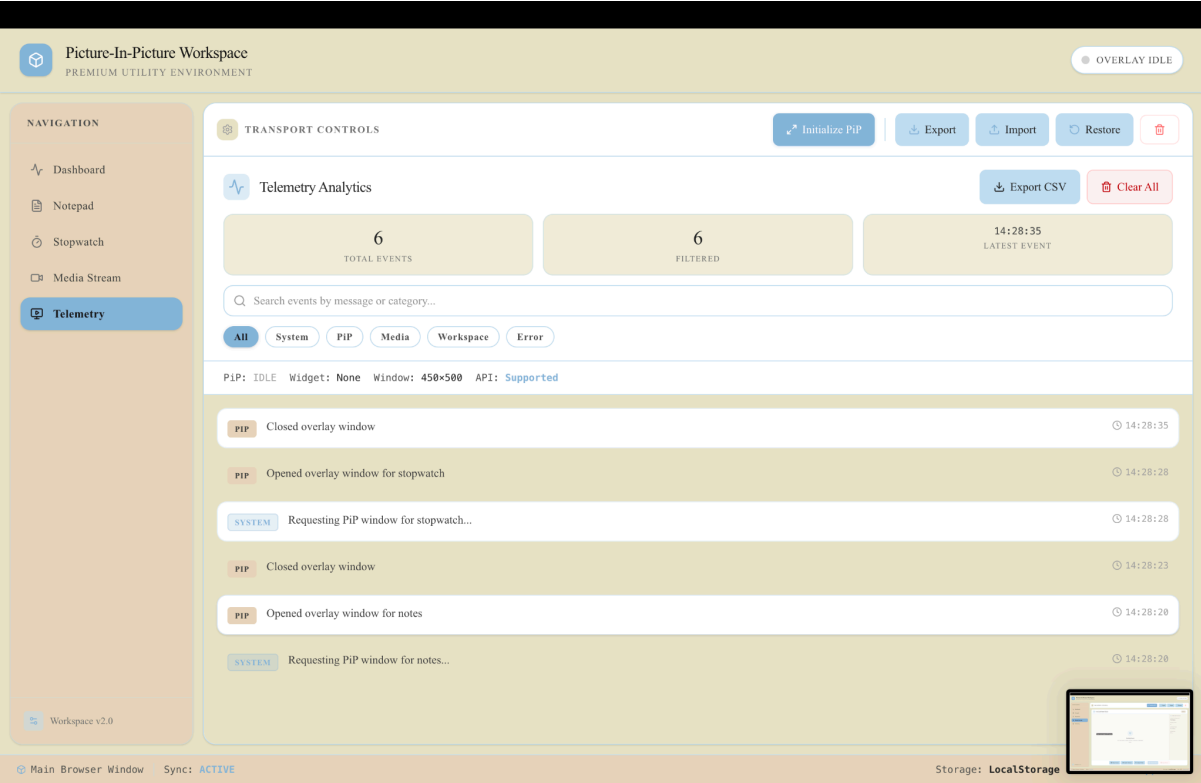
Screenshot 3 – Stopwatch Workspace



Screenshot 4 – Media Stream Viewer



Screenshot 6 – Telemetry Console



Conclusion

The Picture-In-Picture Overlay Workspace Utility successfully demonstrates how modern browser APIs can be used to create advanced productivity tools without requiring backend infrastructure.

The application integrates workspace management, note-taking, media streaming, state synchronization, and floating overlay windows into a unified platform. By leveraging ReactJS, Zustand, Tailwind CSS, LocalStorage, and the Document Picture-in-Picture API, the project provides an efficient multitasking environment while maintaining a lightweight frontend-only architecture.

GitHub Repository

GitHub Link:

<https://github.com/pallavi-676/Picture-in-Picture-caseStudy>

Live Project Link

Live Demo:

<https://picture-in-picture-pallavi-676s-projects.vercel.app/>